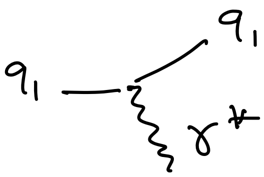
Direct evidence for quarks

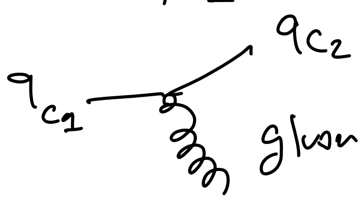
quarks: spin 1/2 particles
small mass

electric charge $Q = Ze$ $\quad Z < 1 \longrightarrow$ interaction w/ photon.

color charge $\longrightarrow$ interaction w/ gluons

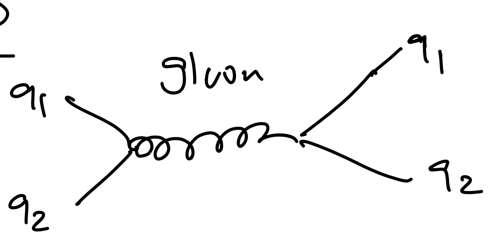
also interact with weak interaction
mediated $W^{\pm}, Z^{0}$

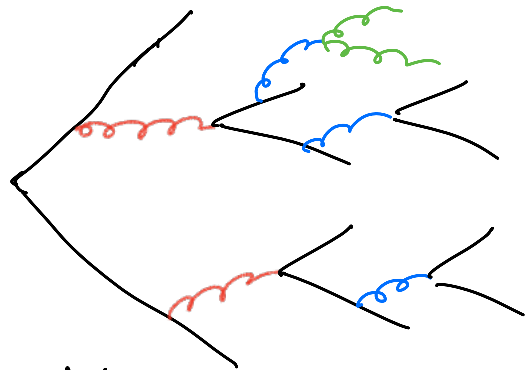




quarks contained in hadrons (mesons, baryons)
free quarks do not exist : color confinement

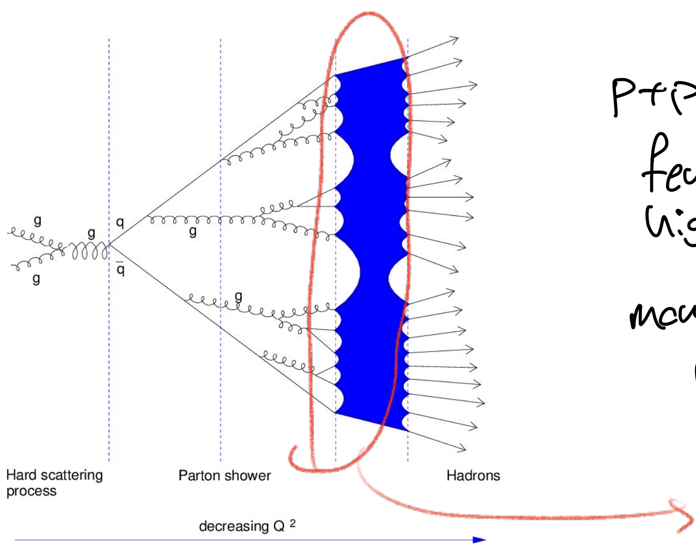
QCD





gluon splitting continues until some energy threshold
$\Rightarrow$ quarks have to form bound states
$\Rightarrow$ observe hadrons.

hadron scattering

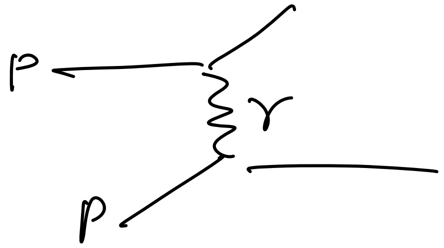


Hadronization

$p + p \longrightarrow$ hadrons
few hadrons with high energy.

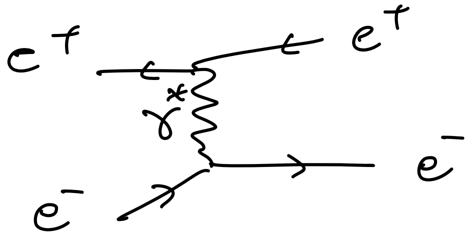
many pions with very small energy.

$p+p \xrightarrow{\text{QED?}} \quad << \quad p+p \xrightarrow{\text{QCD}}$



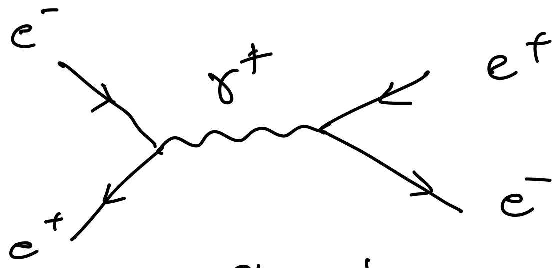
$\Rightarrow$ QED in e^+e^- collisions.

$e^+e^- \longrightarrow e^+e^-$ Bhabha scattering



t-channel

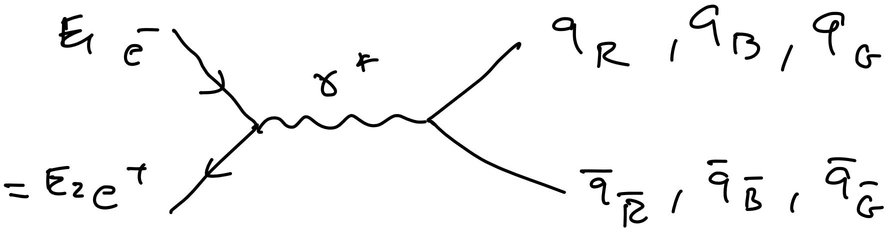
$+$



s-channel
annihilation

$$\mathcal{M}(e^+e^- \to e^+e^-) = \mathcal{M}_t + \mathcal{M}_s$$

$\Rightarrow$ s channel to produce quarks.

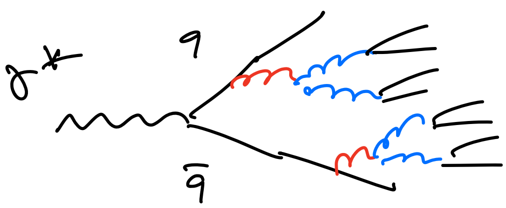


Inelastic scattering

$\underline{P}_1 = (E_1 \vec{P}_1) \qquad \underline{P}_2 = (E_1 - \vec{P}_1)$

$\sqrt{s} = \sqrt{(\underline{P}_1 + \underline{P}_2)^2} = 2E$ total energy available.

$\sqrt{s} \geqslant 2m_q$

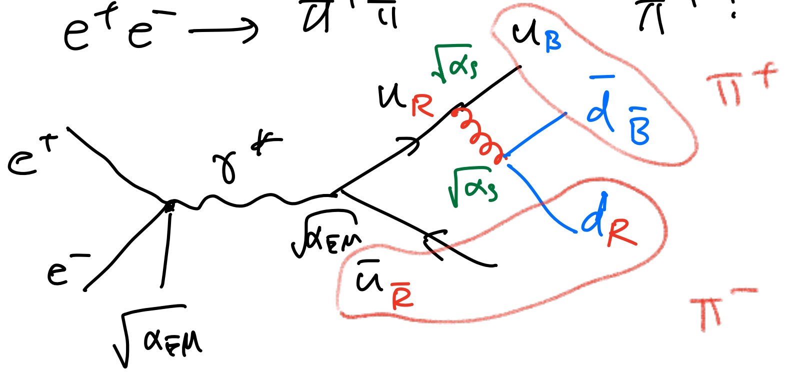


hadrons

$e^+e^- \to \pi^+\pi^-$

$\pi^+ : \; u\bar{d} \qquad \pi^- : \; \bar{u}d$

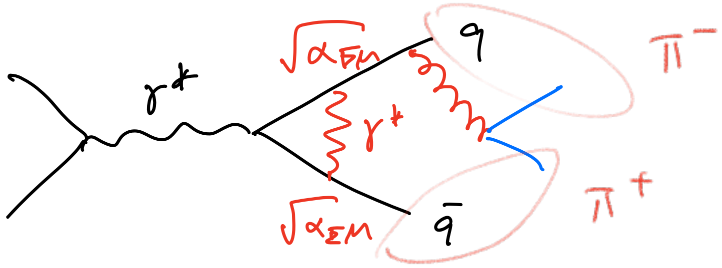
gluon: $\bar{R}B$

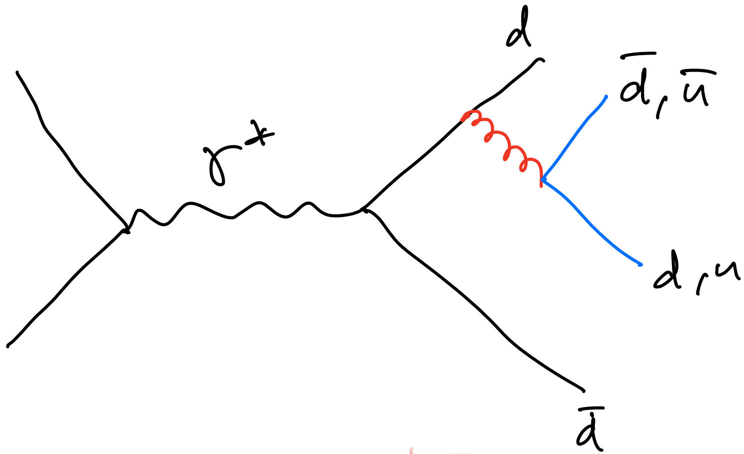


physical process: $e^+e^- \longrightarrow \pi^+\pi^-$

$$2E = \sqrt{s} \geq 2m_\pi \quad \Rightarrow \quad E_{beam} \geq m_\pi = 140\,\text{MeV}$$

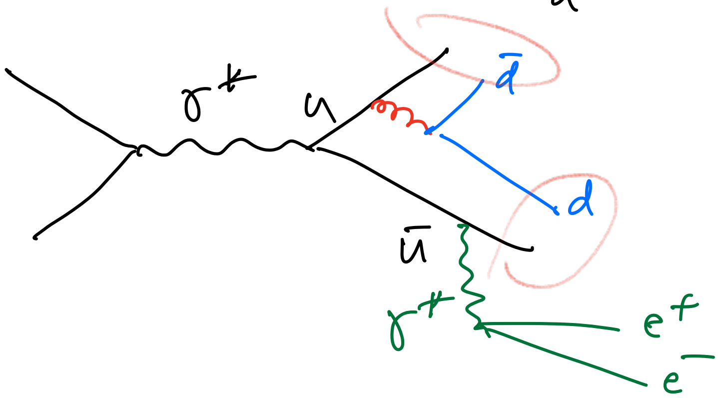
$$\sqrt{s} \equiv 2m_\pi \quad \Rightarrow \quad e^+e^+ \longrightarrow \pi^+\pi^-\pi^0$$





$e^+e^- \longrightarrow \pi^0\pi^0$

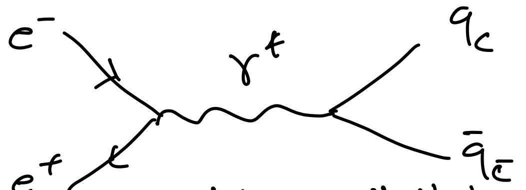
only if $\sqrt{s} \geq 2m_{\pi^0}$



$e^+e^- \longrightarrow \pi^+\pi^- e^+e^-$

$$\sqrt{s} = \underbrace{\underbrace{2m_{\pi^\pm}}_{270\,\text{MeV}} + \underbrace{2m_{e^\pm}}_{1\,\text{MeV}}}_{}$$

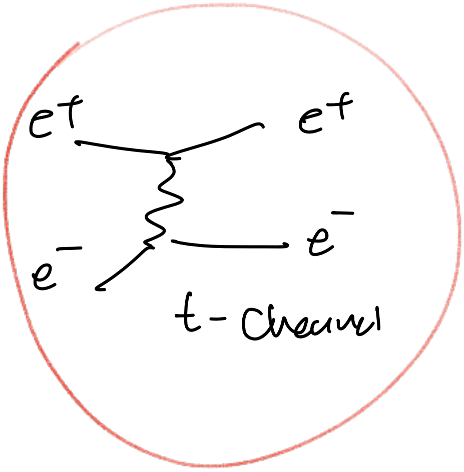
within the experimental resolution of E_{beam}



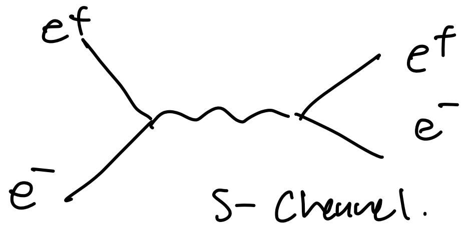
only annihilation.

$\mathcal{M} = ?$ $e^+e^- \to$ hadrons

$e^+e^- \to$ hadrons similar to $e^+e^- \to e^+e^-$ but different

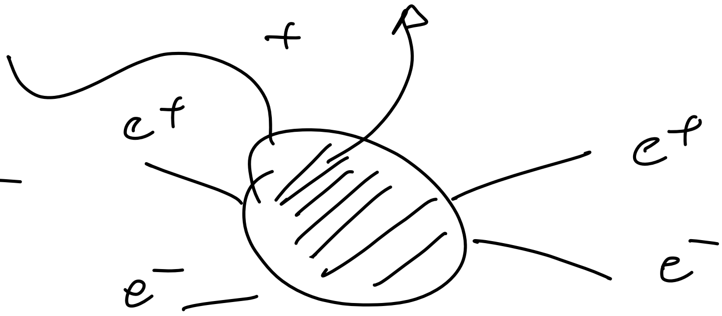


+

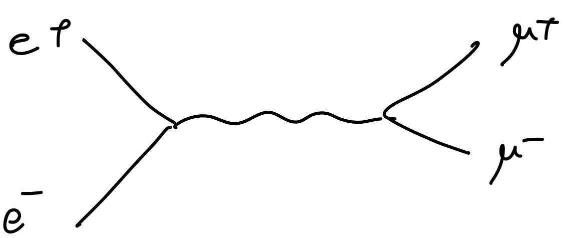


$e^+e^- \to e^+e^-$

+



next lepton: $\mu^\pm$ $m = 106$ MeV $e^+e^- \to \mu^+\mu^-$



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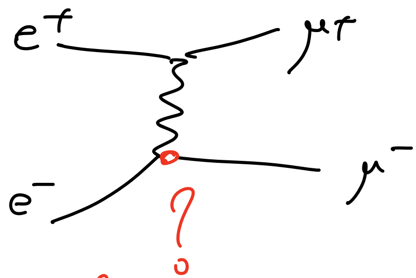
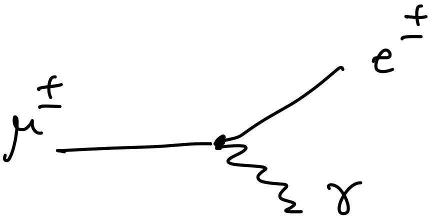


lepton flavor changing.



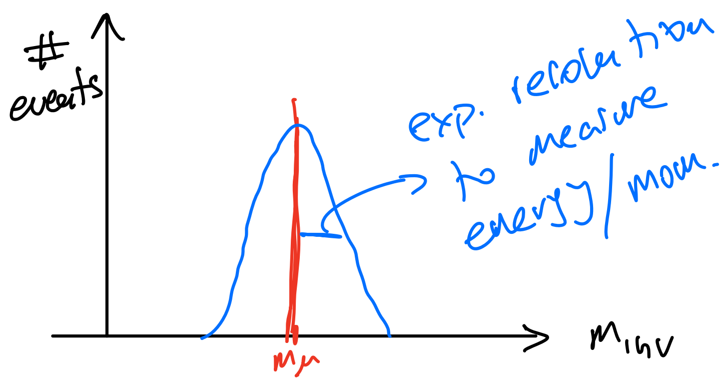
$\mu^- \to e^- + \gamma$

$Q = m_\mu - m_e - m_\gamma \simeq 105.5 \text{ MeV} > 0$

$\gamma \longleftarrow \circ \longrightarrow e^-$ $\Rightarrow$ momentum of e^-, γ

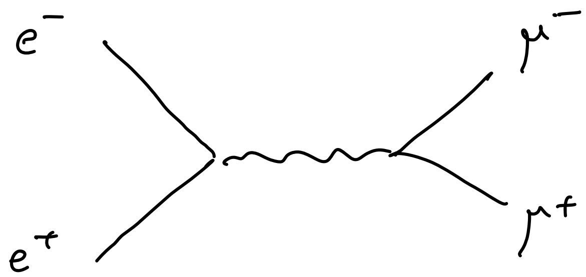
measure $M_{inv} = \sqrt{(\underline{P}_{e^-} + \underline{P}_\gamma)^2}$

MEG experiment @ PSI (Zürich)



$\mathcal{M}(e^+e^- \to q\bar{q}) \sim \mathcal{M}(e^+e^- \to \mu^+\mu^-)$

$\Rightarrow$ compute cross section $\sigma(e^+e^- \to \text{hadrons})$

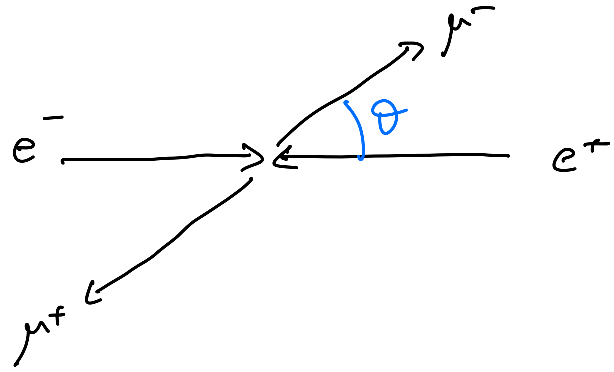


$\underline{P}_3 = (E^-, \vec{P}_{\mu^-})$

$\underline{P}_4 = (E^+, \vec{P}_{\mu^+})$

$\underline{P}_1 = (E, \vec{P}_1)$

$\underline{P}_2 = (E, -\vec{P}_1)$



Center of mass

$\Sigma_i \vec{P}_i = \emptyset$

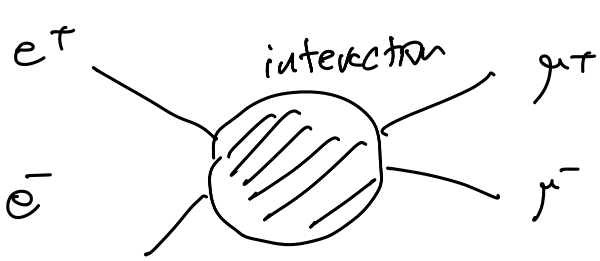
$\Rightarrow \vec{P}_{\mu^+} = -\vec{P}_{\mu^-}$

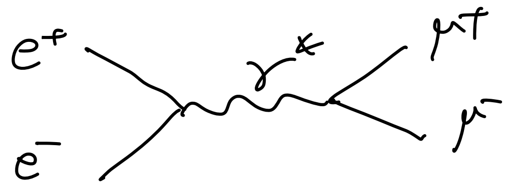
$\sqrt{s} \geq 2m_\mu \Rightarrow 2E \geq 2m_\mu \Rightarrow E_{beam} \geq m_\mu$

if $\sqrt{s} < 2m_\mu$: only $e^+e^- \to e^+e^-$

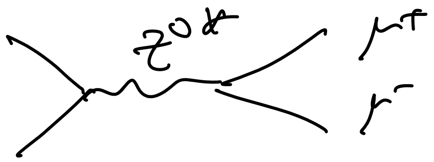
$\sqrt{s} \geq 2m_\mu$: e^+e^-
$\mu^+\mu^-$

if $E_{beam} \gg m_\mu$: e^+e^-
$\mu^+\mu^-$
$q\bar{q}$?

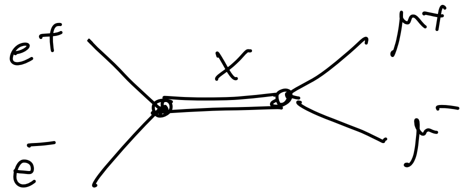




QED

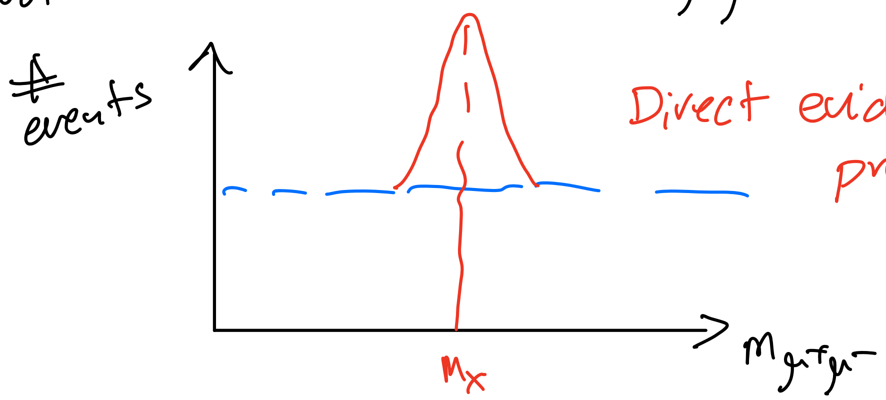


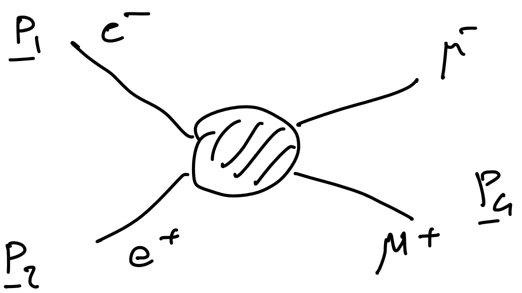
$\sqrt{s} \simeq m_Z \simeq 90$ GeV



$2E = \sqrt{s} \geq m_X \implies E \geq \frac{1}{2} m_X$

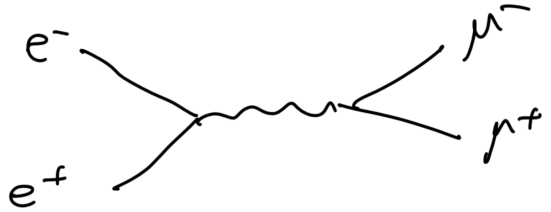
look at invariant mass $m_{\mu^+\mu^-}$ measure $\vec{p}_{\mu^+}$ $\vec{p}_{\mu^-}$





$\vec{P}_{in} \equiv \vec{P}_1$ $\vec{P}_{out} \equiv \vec{P}_3$

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2} \frac{1}{(E_1+E_2)^2} \frac{|\vec{P}_{out}|}{|\vec{P}_{in}|} |M|^2$$



$\sqrt{s} = \sqrt{(\underline{P}_1+\underline{P}_2)^2} = \sqrt{2E}$

$E_1 = E_2 = E$: beam energy. $(E_1+E_2)^2 = (2E)^2 = s$

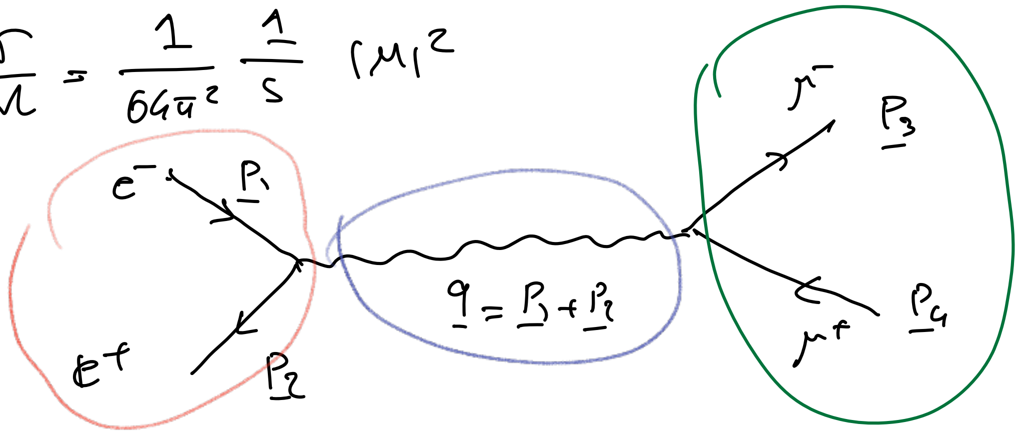
$\vec{P}_{in} = \vec{P}_{e^-}$ $\vec{P}_{out} \equiv \vec{P}_{\mu^-}$

$E >> m_\mu$ $|\vec{P}_e| \simeq E$

$E_\mu = \sqrt{m_\mu^2 + \vec{P}_\mu^2}$

$E >> m_\mu \implies P_\mu >> m_\mu \implies |\vec{P}_\mu| \simeq E$ (massless muon)

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2}\frac{1}{s}|M|^2$$



$$q^2 = s$$

$$-iM = \left[\bar v(p_2)\, ie\gamma^\mu u(p_1)\right] \frac{-ig_{\mu\nu}}{q^2} \left[\bar u(p_3)\, ie\gamma^\nu v(p_4)\right]$$

Spin configurations

Initial states

$e^-_L \xrightarrow{\Leftarrow} \quad \xleftarrow{\Leftarrow} e^+_R \qquad S_z = -1$

$e^-_L \xrightarrow{\Leftarrow} \quad \xleftarrow{\Rightarrow} e^+_L \qquad S_z = 0$

$e^-_R \xrightarrow{\Rightarrow} \quad \xleftarrow{\Leftarrow} e^+_R \qquad S_z = 0$

$e^-_R \xrightarrow{\Rightarrow} \quad \xleftarrow{\Rightarrow} e^+_L \qquad S_z = +1$

Final states:

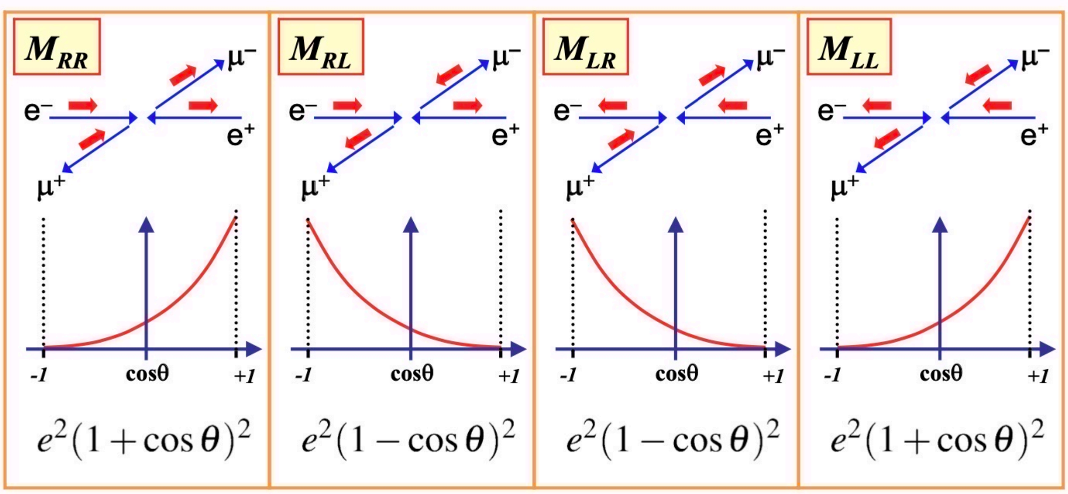
$\mu^-_L \mu^+_R$

$\mu^-_L \mu^+_L$ $\qquad \Rightarrow$ 16 Combinations

$\mu^-_R \mu^+_R$

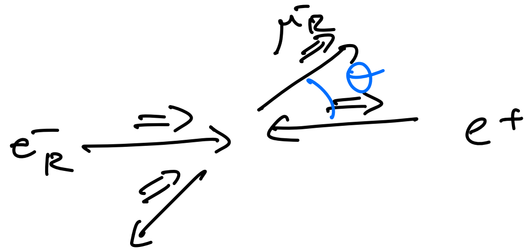
$\mu^-_R \mu^+_L$

only combinations with $S_z = \pm 1$ contribute



M_{AB}: A: e^-_A B: μ^-_B

M_{RR}



$S_z = +1$ $S_z = -1$ $\theta = 0$

$S_z = +1$ $S_z = +1$ $\theta = \pi$

$$|M_{\underset{e^-\mu^-}{RL}}|^2 = |M_{LR}|^2 = (4\pi\alpha)^2(1-\cos\theta)^2$$

$$|M_{RR}|^2 = |M_{LL}|^2 = (4\pi\alpha)^2(1+\cos\theta)^2$$